

Extreme Heat and the Reduction of the South-Facing Premium: Evidence from Shanghai’s Housing Market^{*}

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Abstract

South-facing apartments sell at a price premium in Chinese housing markets, and the hedonic literature treats this premium as a fixed feature value. Using 217,640 resale transactions in Shanghai from 2013 to 2020, we show that it is not fixed. We estimate a within-community difference-in-differences that interacts a south-facing indicator with the extreme-heat days a unit accumulates while it is listed, holding community, quarter, and listing duration fixed. Each additional 100 extreme-heat days lowers the premium by about 0.6 percentage points, from 1.61 percent at no exposure. The adjustment is made on the seller side: it shows up in the listing price but not in the listing-to-transaction discount, which points to sellers who over-weight recent heat rather than to frictionless climate pricing. The south-facing premium is thus endogenous and weather-dependent, and valuation models that treat orientation as a fixed feature will overstate it in hot-summer markets and understate it in cool ones.

Keywords: extreme heat, south-facing premium, hedonic pricing, climate capitalization, Shanghai

JEL codes: Q51, Q54, R31, D91

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1 Introduction

The south-facing orientation is the most desired apartment attribute in Chinese housing markets. It provides longer daily sunlight, passive winter warmth, and a brighter interior (Lu, 2018; Chau et al., 2001). In our sample of Shanghai resale transactions, about 85 percent of units are south-facing. Within the same community, and after controlling for building characteristics, a south-facing unit sells for about 1.6 percent more than an otherwise similar non-south unit.¹ The hedonic literature treats this premium as a fixed component of the characteristic vector (Lu, 2018; Li et al., 2020; Lee, 2024). Yet the same sunlight that buyers value in spring and winter makes a south-facing room hotter and less comfortable when the listing window contains many extreme-heat days. We find that this premium is not fixed: it erodes as extreme-heat days accumulate during the listing window, and we ask through what mechanism it operates and which side of the market drives it.

Recent work documents that extreme heat reduces housing prices in Chinese cities (Kang et al., 2024), that high temperatures raise bid prices in land auctions through buyer cognition (Chen et al., 2023), that climate shocks surface in commercial real estate (Holtermans et al., 2024), and that flood and sea-level risks are capitalized to the extent that buyers believe in the hazard (Baldauf et al., 2020; Bernstein et al., 2019; Giglio et al., 2021). We build on this literature in three ways. First, we show that the south-facing premium, which the hedonic literature treats as a fixed characteristic price, is in fact endogenous: it moves with the extreme-heat exposure a unit accumulates while on the market, so the premium is shaped by weather rather than fixed at the level of a primitive attribute value. Second, we identify this response within community-by-quarter cells, interacting orientation with exposure while holding the local price level fixed; this separates the orientation-specific heat response from city-wide price movements and tells cumulative-exposure capitalization apart from effects that act at the moment of sale. Third, we place the adjustment on the seller side: it is set in the listing price rather than extracted through end-of-window bargaining, which points to extrapolative pricing by sellers who over-weight recent heat rather than a frictionless capitalization of a fixed climate.

Shanghai sits near 31°N. From June through September, daily mean temperatures regularly exceed 28.9°C, which is the 90th percentile of the 2000–2020 daily-mean distribution. On such days, south-facing rooms receive direct solar radiation through the morning and early afternoon, raising indoor temperatures above those in north-facing rooms in the same

1. Estimated from the baseline regression in Section 5. In levels, south-facing and non-south units sell at similar average prices, since south-facing units span the full range of building quality; the premium appears once units are compared within the same community.

building. Older walk-up buildings, with thin walls and single-pane windows, amplify the gap. When extreme heat recurs during the listing window, the summer discomfort of a south-facing unit becomes more salient to the buyers and sellers who price it.

We use 217,640 transactions across 8,793 residential communities, drawn from a major Chinese real-estate platform between 2013 and 2020. For each transaction we compute the number of extreme-heat days during the listing window, defined as the interval between the listing date and the transaction date. The mean exposure is 47.8 days, the median is 34 days, the 75th percentile is 88 days, and the maximum is 260 days. Our regression follows the hedonic tradition ([Rosen, 1974](#)) with thirty building controls (five transaction-class and usage indicators are collinear with the community fixed effects and drop out, leaving twenty-five estimated), community fixed effects, year-by-quarter fixed effects, and listing-duration-group fixed effects. Standard errors are clustered two-way by community and transaction date.

We find that the south-facing premium falls as extreme heat accumulates during the listing window. At zero exposure the premium is 1.61 percent. Each additional 100 extreme-heat days lowers it by about 0.6 percentage points, so the premium falls to 1.33 percent at the sample mean exposure of 48 days, to 1.09 percent at the 75th percentile of 88 days, and close to zero at the maximum of 260 days. The estimate is significant at the one percent level.

Several further results sharpen this finding. The response is non-linear in cumulative exposure: in an event-study specification with cumulative-heat bins, in the spirit of [Deschênes and Greenstone \(2011\)](#), the interaction is near zero in the first thirty days, turns sharply negative once the count passes thirty days, and attenuates in very long windows. The channel is thermal comfort, since triple interactions with listing-period humidity, wind, and cloud cover move the penalty in the directions comfort physics predicts, with humidity amplifying it and wind and cloud cover offsetting it. The adjustment is made on the seller side: the penalty appears in the listing price at nearly the same size as in the transaction price, while the listing-to-transaction discount does not move, in line with the seller-side pricing in [Genesove and Mayer \(2001\)](#). The size of the effect depends on how much of the listing window falls inside one summer, and in a single-summer subsample, where the cumulative count is not diluted by cool months from other seasons, the interaction is about ten times the baseline.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the data and variables. Section 4 presents the specification. Section 5 reports baseline, event-study, and event-indicator results. Section 6 examines mechanism. Section 7 reports robustness checks. Section 8 translates the estimate into its welfare magnitude. Section 9 concludes.

2 Literature Review

The south-facing premium and hedonic pricing. The hedonic framework of [Rosen \(1974\)](#) treats a house as a bundle of characteristics and its price as the market value of that bundle. Within this framework a large literature prices individual attributes. [Chau et al. \(2001\)](#) find that culturally favored features carry measurable premiums in Hong Kong apartments. For orientation, [Lu \(2018\)](#) estimates the south-facing premium in Shanghai at about 14 percent, [Li et al. \(2020\)](#) report about 1.4 percent in Chongqing, and [Lee \(2024\)](#) finds a premium that ranges from near zero to about 13 percent across Korean regions. The physical basis for valuing orientation is documented in the thermal-comfort literature: [Chen et al. \(2015\)](#) show in a Shanghai field survey that air temperature and solar radiation are the two dominant predictors of perceived thermal sensation, and [Lin and Deng \(2008\)](#) map temperature, humidity, and bedding into a comfort score for subtropical sleeping environments. A closely related strand prices thermal performance directly. [Lu and Spaenjers \(2025\)](#) show with French data that house prices respond to coarse energy labels rather than to continuous energy use, with price drops of between 1.9 and 4.7 percent across adjacent label thresholds, which they read as evidence that buyers and sellers price simplified summaries rather than fundamentals. [Brounen and Kok \(2011\)](#) and [Aydin et al. \(2020\)](#) find premiums on the order of 1 to 5 percent for higher energy ratings in Europe.

This work treats the orientation premium as a fixed characteristic price, estimated as a single coefficient that does not vary with conditions. The closest study to ours is [Lu and Spaenjers \(2025\)](#), who show that the market prices a coarse label rather than the continuous quantity behind it. We extend this literature in two ways. First, we show that the south-facing premium is state-dependent rather than fixed: within the same community and quarter, it contracts as extreme-heat days accumulate during the listing window, so the premium behaves as an endogenous object shaped by weather rather than a primitive attribute value. Second, where [Lu and Spaenjers \(2025\)](#) document heuristic pricing of a static label, we identify a case in which a salient recent experience, namely accumulated heat during the window, moves the price of one attribute, and we trace the response to the seller side.

Extreme weather and housing markets. A separate literature studies how temperature and climate exposure affect prices and behavior. At the aggregate level, [Burke et al. \(2015\)](#) document a strongly non-linear relationship between temperature and economic output. [Deschênes and Greenstone \(2011\)](#) introduced the temperature-bin approach and find that days with a mean temperature above 90°F raise U.S. age-adjusted mortality by about 0.11 percent, and [Park et al. \(2021\)](#) show that hot days raise workplace injury and

lower productivity. In housing, [Kang et al. \(2024\)](#) use 2009-to-2019 data on 283 Chinese prefecture-level cities and find that one additional day above 35°C in a year is associated with about a 0.1 percent decline in city-level prices, with a similar negative effect for a heatwave indicator of three or more consecutive days above 35°C. [Chen et al. \(2025\)](#) report that heatwave intensity reduces real-estate prices in China between 2010 and 2020, with stronger responses in inland and fiscally stressed cities. At finer resolution, [Chen et al. \(2023\)](#) find that an additional extreme-heat day in the fifteen days before a land auction raises the average bid by 0.6 percent, which they attribute to heat-induced cognitive impairment, and [Pan et al. \(2022\)](#) report a 7.1 percent winter price discount in southern Chinese cities that lack district heating. Adverse weather also changes the pace of trade: [Cascarano and Natoli \(2023\)](#) find with Italian listing data that hot days slow search and reduce transaction volume, and [Holtermans et al. \(2024\)](#) document a discount on commercial properties more exposed to climate shocks.

Whether climate exposure is capitalized depends on what buyers know and believe. [Baldauf et al. \(2020\)](#) find that homes in U.S. flood zones sell at a discount only where residents accept climate change, and [Bernstein et al. \(2019\)](#) find a discount of about 7 percent for sea-level-rise exposure that is concentrated among sophisticated buyers. [Bakkensen and Barrage \(2022\)](#) pair a Rhode Island coastal survey with a structural model and show that under-perception of flood risk leaves coastal prices 6 to 13 percent above fundamentals. [Ortega and Taşpınar \(2018\)](#) use Hurricane Sandy to identify persistent declines in newly recognized flood zones, [Grühl et al. \(2025\)](#) find with the 2021 Western European flood that prices fall mainly near inundated areas and where climate beliefs are stronger, and [Fairweather et al. \(2024\)](#) show in a field experiment that delivering flood-risk information changes how buyers search and bid. [Giglio et al. \(2021\)](#) recover market-implied discount rates for climate risk from long-leasehold prices, and [Severen et al. \(2018\)](#) show that U.S. farmland already prices in climate forecasts, so that a naive Ricardian valuation overstates damages by 50 to 62 percent. A third group studies weather as a short-lived shock at the moment of sale. [Hu and Lee \(2020\)](#) find that sunshine and temperature on the day of a Sydney auction raise the sale price, with the effect concentrated where bidding competition amplifies sentiment, [Busse et al. \(2015\)](#) show that weather on the day of car shopping changes purchase choices, and [Qin et al. \(2019\)](#) and [Huang and Du \(2021\)](#) find that air pollution on the day of a deal moves housing and land prices. [Genesove and Han \(2012\)](#) provide the search-and-matching framework that links time on the market to price formation.

We extend this literature in three ways. First, we work at the level of a single unit and a single physical attribute rather than at the city-year scale of [Kang et al. \(2024\)](#) or the fixed deal-window of [Chen et al. \(2023\)](#), which lets us read the response from the contrast

between south and non-south units in the same community and quarter. Second, our window of interest is the whole listing period rather than the day of sale, so we can separate the effect of cumulative exposure, which builds up while a unit is on the market, from the deal-day sentiment effects studied by [Hu and Lee \(2020\)](#) and [Qin et al. \(2019\)](#). Third, we place the adjustment on the seller side, in the listing price rather than in end-of-window bargaining, which points to extrapolative pricing rather than a frictionless capitalization of a fixed climate.

3 Data and Variables

3.1 Institutional background

In Shanghai, resale apartments are typically listed and sold through brokerage platforms. A seller who wishes to dispose of a property posts a listing on the platform with photos, an asking price, and a standardized feature sheet that includes orientation, total floor count, the unit’s floor, building structure, and decoration status. The orientation field is one of the most prominently displayed attributes because Chinese buyers and brokers regard orientation, alongside school district and floor level, as a primary screening criterion. Buyers browse listings online, request showings through the platform, visit the property in the company of an agent, and after one or more visits enter price negotiation. When buyer and seller agree, a contract is signed and the transaction is registered. The listing date and transaction date that we observe correspond to the platform post date and the contract date respectively; the interval between them is the window during which the buyer can experience the weather inside the unit. Because most buyers visit a shortlisted property more than once before deciding, the cumulative count of hot days during this window is a natural proxy for the buyer’s stock of indoor heat experience.

3.2 Data sources

The transaction data come from a major Chinese real-estate platform and cover 264,400 second-hand apartment sales closed in Shanghai between 2013 and 2021, with listing dates beginning in 2012. Each record contains the transaction price and date, listing price and date, building area, orientation, floor level, building type, building structure, decoration level, number of rooms, halls, kitchens, and bathrooms, elevator availability, year built, and other structural attributes. The platform is the dominant brokerage for second-hand transactions in Shanghai and is generally considered to cover the universe of brokered resale

activity in the city during this window; sales transacted directly between owners without going through any platform are not in the data, but these are a small fraction of activity in tier-one Chinese cities.

The community data cover 12,495 residential communities in Shanghai. Variables include floor-area ratio, greening ratio, property-management fee, school-district status, and number of buildings. We merge community-level information with the transactions at the community name level and match more than 99 percent of records. The community variables matter for descriptive comparisons but enter the regression only through the community fixed effect, which absorbs any time-invariant community attribute.

The weather data are daily station observations for Shanghai between January 2000 and December 2020, a total of 7,671 days. The variables are daily mean temperature, daily precipitation, daily mean relative humidity, daily mean wind speed, and daily sunshine duration. The 90th percentile of daily mean temperature over this twenty-one-year window is 28.9°C, and the percentile of interest is essentially identical when computed only over the years that contribute transactions to our sample (2012 to 2020). The weather variables are constructed at the daily level so that we can sum them within each listing window without losing information, and the cumulative count we use is independent of how the listing window happens to be split into months or quarters.

We keep transactions with valid listing and transaction dates, a matched community identifier, and complete daily weather coverage across the listing window. The weather record ends in December 2020, so the analysis sample covers transactions through 2020 even though the raw data extend into 2021. This yields 230,783 records used for descriptive statistics. The baseline regression sample further drops 1,749 singleton observations absorbed by community fixed effects and records with missing values on any of the building controls (building area is missing for about 4,600 records, building age for about 6,200). The final sample contains 217,640 transactions in 8,793 communities. Source files are described in the data appendix.

3.3 Variable definitions

Outcome. The dependent variable is the unit transaction price $P_{i,t}$, measured in ten-thousand yuan per square meter and entered in logs. The sample mean price is about 44,000 yuan per square meter, so $\ln P_{i,t}$ has a mean of 1.49.

South-facing dummy. A dummy equal to one if the recorded orientation is "south" or "south-north," following [Lu \(2018\)](#). About 84.7 percent of the sample carries the south-

facing indicator, of which the great majority (about 91 percent of the south-facing units, or 77 percent of the full sample) are due south. The high share is a long-run supply-side equilibrium rather than buyer revealed preference at the margin: Shanghai developers orient new buildings to maximise the number of south-facing units precisely because that orientation carries the largest premium, so the observed share already capitalises the average willingness to pay. What we estimate is not the level of the premium but how that within-community premium responds to listing-window weather.

Cumulative extreme-heat exposure. For each transaction i with listing date l_i and transaction date t_i , we count the days within $[l_i, t_i]$ on which the daily mean temperature exceeds the 90th percentile threshold T^{P90} computed from the 2000 to 2020 station record:

$$\text{Heat}_{i,t} = \frac{1}{100} \sum_{d \in [l_i, t_i]} \mathbb{1}(T_d^{\text{mean}} > T^{P90}). \quad (1)$$

We divide by one hundred so that the coefficient is expressed per one hundred extreme-heat days. The sample mean is 0.478 (47.8 days), the median is 0.34, the 75th percentile is 0.88, the standard deviation is 0.526, and the maximum is 2.60.

Modifiers. Listing-period averages of daily mean humidity, daily mean wind speed, and daily cumulative sunshine, plus counts of days above the 90th percentile of humidity, wind speed, and below the 10th percentile of sunshine (proxy for cloudy days).

Controls. Thirty building attributes (the five transaction-class and usage indicators are collinear with the community fixed effects and drop out, leaving twenty-five estimated): building area and its square; building age and its square; counts of rooms, halls, kitchens, and bathrooms; dummies for high floor and middle floor (low floor is the omitted category); elevator availability; decoration level (renovated, semi-renovated, raw, with the residual category omitted); building type (slab, tower, mixed); building structure (steel and concrete, brick and concrete, frame); apartment layout (flat, duplex, multi-floor); property rights duration (70-year, 50-year); and transaction class (commodity, affordable, relocation; ordinary residential, villa).

3.4 Descriptive statistics

Table 1 reports summary statistics for the main variables. Table 2 compares south-facing and non-south units on price and on the main characteristics, as a check on how similar the

two groups are before the regression.

TABLE 1 Summary statistics

Variables	Obs	Mean	SD	p25	p75
<i>Panel A. Outcome</i>					
ln unit price (10,000 yuan/m ²)	230,783	1.49	0.42	1.22	1.77
<i>Panel B. Orientation</i>					
South-facing dummy	230,783	0.847	0.360	1.00	1.00
<i>Panel C. Heat exposure and listing window</i>					
Extreme-heat days (cumulative)	230,783	47.82	52.58	0	88
Listing duration (days)	230,783	399.22	429.72	56	625
<i>Panel D. Building characteristics</i>					
Building area	226,188	79.00	34.09	29.50	216.58
Building age	224,550	18.72	9.82	0.00	98.00
Bedrooms	229,763	1.97	0.82	0.00	9.00
Living rooms	229,763	1.35	0.63	0.00	8.00
Bathrooms	229,763	1.18	0.47	0.00	7.00

Notes: Sample is Shanghai resale transactions from 2013 to 2020 with full data. The dependent variable is the log unit transaction price (mean 1.49, about 44,000 yuan per square meter). Panel C reports the cumulative extreme-heat measure (days above the 90th-percentile daily-mean temperature within the listing window) and the listing duration over which it accumulates; the heat measure is zero at p25 because more than a quarter of listing windows fall entirely outside the local summer. The full set of building controls and listing-window weather modifiers used in the regressions is described in Section 3. N varies across rows because building area and building age are missing for a small share of records.

The two groups are balanced on the building characteristics that carry hedonic value: average building area differs by about a quarter of a square meter and building age by about a quarter of a year, neither significant. The unit price in levels is almost identical across the two groups, which is why the south-facing premium appears only once units are compared within the same community rather than across the whole market. Two differences are worth noting. South-facing units carry fewer cumulative extreme-heat days and shorter listing windows, which follows from the fact that they sell faster, so their windows are shorter and accumulate less heat; the listing-duration fixed effects together with the within-community comparison hold this difference fixed, and Section 7.2 probes it directly. South-facing units are also somewhat larger, with slightly more rooms on average, which the building-size controls absorb.

TABLE 2 Summary statistics by orientation: south-facing versus non-south

Variable	(1)South-facing		(2)Non-south		(1) - (2)
	Mean	SD	Mean	SD	Difference
ln unit price	1.494	0.409	1.473	0.453	0.020**
Unit price (10,000 yuan/m ²)	4.797	1.835	4.799	1.999	−0.002
Extreme-heat days (cumulative)	44.25	51.17	67.51	55.82	−23.26***
Listing duration (days)	369.4	415.5	557.0	470.5	−187.58***
Building area (m ²)	79.04	33.75	78.78	35.96	0.26
Building age (years)	18.67	9.71	18.96	10.43	−0.28
Bedrooms	2.018	0.785	1.697	0.941	0.321***
Living rooms	1.386	0.609	1.163	0.731	0.223***
Bathrooms	1.193	0.452	1.076	0.568	0.117***

Notes: The sample is the descriptive sample of Shanghai resale transactions from 2013 to 2020; south-facing units are about 85 percent of observations (roughly 195,000 south-facing and 35,000 non-south), and the count varies slightly across rows because building area and building age are missing for a small share of records. “Difference” is the south-facing mean minus the non-south mean, with stars from a regression of each variable on the south-facing indicator with standard errors two-way clustered by community and transaction date. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4 Empirical Strategy

The baseline model is

$$\ln(P_{i,t}) = \alpha + \beta_1 \cdot \text{South}_i + \beta_2 \cdot \text{Heat}_{i,t} + \beta_3 \cdot \text{South}_i \times \text{Heat}_{i,t} + X_i' \gamma + \delta_c + \delta_q + \delta_d + \varepsilon_{i,t}, \quad (2)$$

where X_i is the vector of thirty building controls (twenty-five estimated after five are absorbed by the community fixed effects), δ_c is a community fixed effect (8,793 categories), δ_q is a year-by-quarter fixed effect (32 categories), and δ_d is a listing-duration-group fixed effect (five bands: up to 60 days, 61 to 180, 181 to 365, 366 to 730, more than 730). Standard errors are clustered two-way by community and transaction date.

β_3 is the coefficient of interest. If the south-facing premium is invariant to listing-period heat, $\beta_3 = 0$; if listing-period heat erodes the premium, $\beta_3 < 0$. Here β_1 is the premium at zero exposure and β_3 is how much one hundred P90 days erode it. Section 5 reports both, together with the implied premium $\beta_1 + \beta_3 \cdot H$ at the sample mean, the seventy-fifth percentile, and the maximum of H . The interpretation does not require that β_1 be a structural primitive, only that the within-community contrast hold the local averaging constant.

TABLE 3 Baseline: south-facing premium and extreme-heat exposure

Dependent variable: $\ln(\text{unit price})$	(1) No FE	(2) + Comm.	(3) + Yr $\times$ Qtr	(4) + Duration	(5) Baseline	(6) Comm. cluster
South-facing (β_1)	0.1176*** (0.0117)	0.0215*** (0.0018)	0.0188*** (0.0018)	0.0170*** (0.0018)	0.0161*** (0.0017)	0.0161*** (0.0017)
Heat per 100 days (β_2)	0.1695*** (0.0122)	0.0051** (0.0025)	-0.0143*** (0.0037)	-0.0155*** (0.0037)	-0.0133*** (0.0037)	-0.0133*** (0.0025)
South $\times$ Heat (β_3)	-0.0962*** (0.0100)	-0.0133*** (0.0019)	-0.0099*** (0.0018)	-0.0058*** (0.0018)	-0.0059*** (0.0018)	-0.0059*** (0.0017)
Building controls	No	No	No	No	Yes (30)	Yes (30)
Community FE	No	Yes	Yes	Yes	Yes	Yes
Year-by-quarter FE	No	No	Yes	Yes	Yes	Yes
Listing-duration FE	No	No	No	Yes	Yes	Yes
Clustering	2-way	2-way	2-way	2-way	2-way	Comm.
Observations	217,640	217,640	217,640	217,640	217,640	217,640
R^2	0.0149	0.933	0.934	0.942	0.943	0.943
Adjusted R^2	0.0149	0.930	0.931	0.940	0.941	0.941

Notes: The dependent variable is the log unit transaction price, with a sample mean of 1.49 (about 44,000 yuan per square meter). The coefficients therefore read as proportional effects: in column 5, β_1 is a 1.61 percent south-facing premium at zero heat exposure, and β_3 means that 100 extreme-heat days during the listing window lower that premium by about 0.59 percentage points. Column 1 includes no fixed effects; columns 2 to 4 add community, year-by-quarter, and listing-duration-group fixed effects in turn; column 5 adds the full set of building controls and is the baseline; column 6 repeats the baseline with one-way clustering on community. Building controls comprise building area and its square, building age and its square, room, hall, kitchen, and bathroom counts, high- and middle-floor dummies, elevator availability, decoration level, building type, building structure, apartment layout, and property-rights duration; five collinear transaction-class and usage indicators are absorbed by the community fixed effects. The interaction attenuates as location and duration variation is absorbed, then stabilizes near -0.0059 and is unaffected by the clustering choice. Standard errors in parentheses, two-way clustered by community and transaction date unless noted. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Identification. The community fixed effect absorbs any time-invariant difference between communities, including average orientation mix and average building quality. The year-by-quarter effect absorbs trends in the housing market and seasonal patterns at the city level. The listing-duration-group effect absorbs mechanical correlation between extreme-heat exposure and the length of the listing window. Twenty-five estimated building controls absorb within-community variation in size, age, layout, decoration, and the like. The remaining five transaction-class and usage indicators are absorbed by the community fixed effects. The remaining variation in $\text{Heat}_{i,t}$ is the weather realized during each unit’s particular listing window. As long as this variation is not correlated with unobserved heat-sensitive features of south-facing units in the same community within the same quarter, the interaction coefficient identifies the differential pricing of orientation under heat exposure.

The within-community comparison is the source of identifying variation, so its content is worth spelling out. A Shanghai community is about the smallest unit at which housing in the city is meaningfully homogenous: units share land, building developer, construction year, common amenities, school district, and most observable neighbourhood characteristics. Two units in the same community typically differ only in floor level, exact size, and orientation. The interaction we estimate therefore contrasts south-facing units with non-south units in the same community that transact in the same quarter and have a listing window of comparable length, holding the average market context fixed. The only systematic difference between these units that could drive a differential response to listing-period heat is the physical exposure to sunlight that orientation creates. Any remaining unobserved heterogeneity would have to be both correlated with orientation within a community-quarter cell and differentially responsive to weather, a combination that is hard to construct as a confound.

The main threats are selection (some units may delist during hot months) and omitted variables tied to listing duration. We address selection in Section 7 with auxiliary regressions of the community-quarter transaction count and south-facing share on the community-quarter average heat measure. Within community-quarter cells both coefficients are positive and significant, the opposite sign of what selection-driven dropouts of south-facing units would predict. We address listing duration with the fixed-effect ladder (5, 10, 20, 40 bands), with a heat-share specification, and with subsample analyses described in Section 7.

Specification choices. The heat measure is scaled by one hundred days so that the coefficient is expressed as the change in log price for an additional one hundred days of P90 exposure within a window. This is a natural unit because a Shanghai summer spans roughly one hundred days from late June to early October and the empirical distribution of cumulative heat is concentrated below this value (the 75th percentile is 88 days). Coefficients

expressed per single day would be more than two orders of magnitude smaller and harder to compare with the city-level estimates in Kang et al. (2024) or the deal-window estimates in Chen et al. (2023). Second, we use community fixed effects rather than the coarser sub-district effects of Lu (2018) because a typical Shanghai community is a contiguous group of buildings developed by the same developer at the same time. Within-community variation in orientation is therefore between physically similar units that share land, building stock, and amenities, which is the natural control for unobserved location quality. Third, we cluster standard errors two-way by community and transaction date. Community clustering absorbs the within-community correlation across sales of similar units; transaction-date clustering absorbs city-wide shocks (policy, news, market conditions) that move all prices on a given day. The data has 8,793 communities and 2,837 distinct transaction dates, well within the asymptotic requirements of two-way clustering.

5 Main Results

5.1 Baseline

Table 3 reports the baseline. The estimate of β_3 is -0.00592 with a clustered standard error of 0.00176 ($t = -3.36$, $p = 0.001$). The estimate of β_1 is $+0.01610$ ($t = 9.31$). Writing the implied south-facing premium at exposure level H as

$$\Pi(H) \equiv \mathbb{E}[\ln P_{\text{south}} - \ln P_{\text{non-south}} \mid \text{Heat} = H] = \beta_1 + \beta_3 \cdot H, \quad (3)$$

the no-exposure premium is $\Pi(0) = 1.61$ percent. At the sample mean of 47.82 days, $\Pi(0.4782) = 0.01610 + (-0.00592) \times 0.4782 = 0.01327$, or 1.33 percent. At 88 days (the 75th percentile of exposure) it is 1.09 percent. At 260 days (the maximum) it falls to about 0.07 percent. The break-even level of cumulative heat at which the orientation premium vanishes is

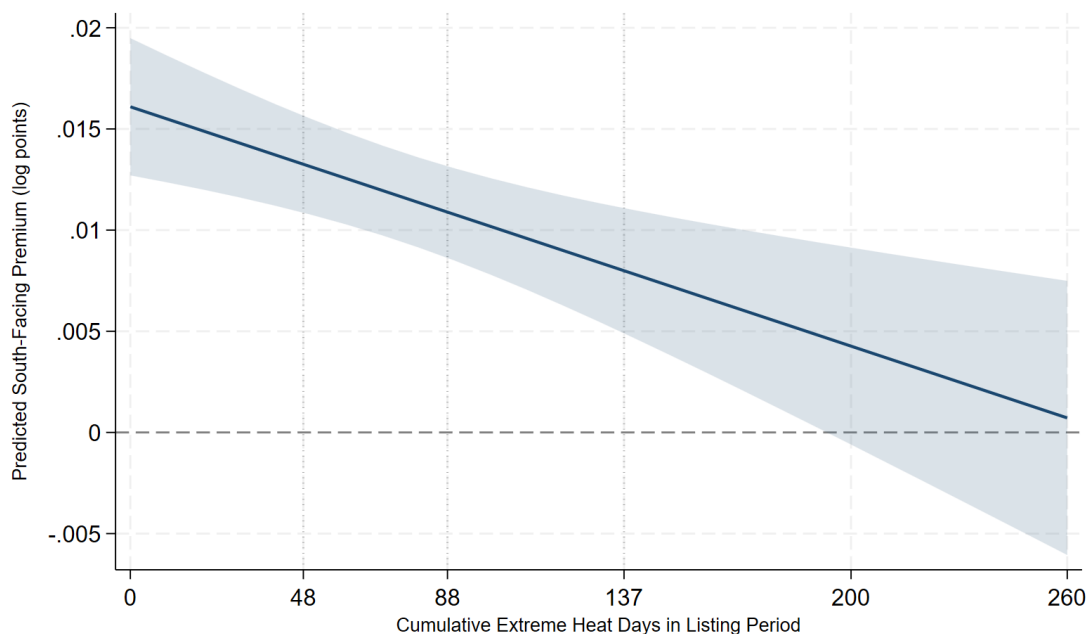
$$H^* = -\beta_1/\beta_3 = 0.01610/0.00592 \approx 2.72, \quad (4)$$

which corresponds to roughly 272 cumulative extreme-heat days, just past the largest exposure we observe (260 days). The premium therefore stays positive across the whole sample and only falls close to zero in the most heat-exposed windows.

The 1.61 percent baseline premium is smaller than the 14 percent that Lu (2018) reports for Shanghai. The difference reflects the finer fixed-effect partition we use. Lu (2018) controls for subdistrict effects and identifies the premium from contrasts between communities within the same subdistrict; we control for community effects and identify the premium from

contrasts within the same community. Within-community variation removes a large share of the cross-sectional component of orientation pricing that Lu (2018)’s subdistrict design retains. The two numbers answer different questions and need not agree. Our interest is in within-community variation that moves with weather, which the community fixed effect preserves.

Figure 1 traces the implied south-facing premium under the linear specification across the empirical range of cumulative heat days. The premium starts at 1.61 percent with zero exposure, falls to about 1 percent at the 75th percentile (88 days), and approaches zero at the maximum (260 days). The shaded band is the 95 percent confidence interval on the linear combination $\beta_1 + \beta_3 \cdot \text{Heat}$.



Notes: Predicted south-facing premium $\beta_1 + \beta_3 \cdot \text{Heat}$ from the baseline regression, plotted across cumulative extreme-heat days in the listing window. Shaded band is the 95 percent confidence interval.

Implied premium computed from the baseline coefficients in Table 3.

FIGURE 1 Implied south-facing premium across the heat range

5.2 Event study

We replace the linear cumulative measure with seven non-overlapping bins on the count of extreme-heat days within the listing window: zero days (bin 0, the reference), one to ten days (bin 1), eleven to thirty (bin 2), thirty-one to sixty (bin 3), sixty-one to one hundred (bin 4), one hundred and one to one hundred and fifty (bin 5), and one hundred and fifty-one

or more (bin 6). The bin structure follows the temperature-range logic of [Deschênes and Greenstone \(2011\)](#). Letting $B_{k,i}$ be the indicator that observation i falls in bin k , we estimate

$$\ln(P_{i,t}) = \alpha + \beta_1 \cdot \text{South}_i + \sum_{k=1}^6 \gamma_k \cdot B_{k,i} + \sum_{k=1}^6 \theta_k \cdot \text{South}_i \times B_{k,i} + X'_i \gamma + \delta_c + \delta_q + \delta_d + \varepsilon_{i,t}. \quad (5)$$

Each θ_k is the differential south-facing premium in bin k relative to bin 0. Figure 2 plots the $\hat{\theta}_k$ with 95 percent confidence intervals. The interaction is close to zero for bins 1 and 2 ($\hat{\theta}_1 = -0.0002$, $\hat{\theta}_2 = -0.0052$, both insignificant), turns sharply negative for bin 3 (thirty-one to sixty days, $\hat{\theta}_3 = -0.0186$, $p < 0.01$), and stays significantly negative but attenuates across bins 4 to 6 ($\hat{\theta}_4 = -0.0164$, $\hat{\theta}_5 = -0.0139$, $\hat{\theta}_6 = -0.0123$). A linear-trend test that replaces the bin dummies with the south-by-bin-midpoint interaction gives a negative and significant slope (-0.0000667 , $p < 0.01$).

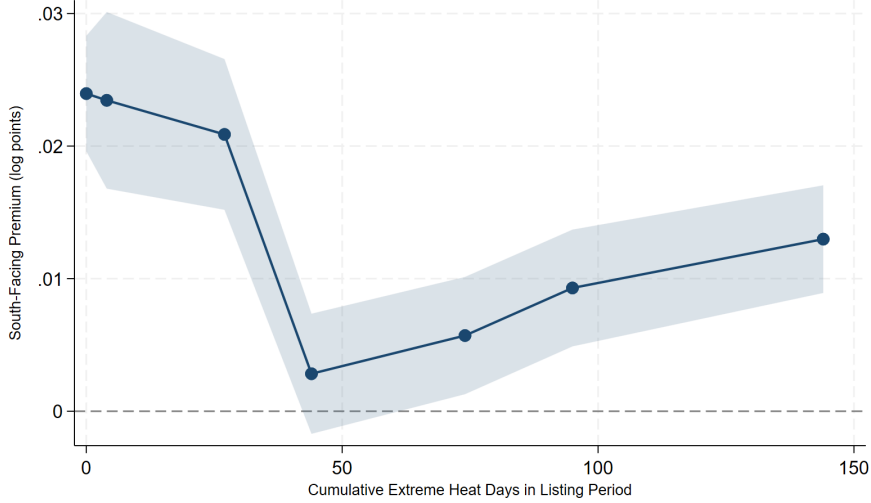
The shape has two features that matter for the interpretation. First, the kink between bin 2 (eleven to thirty days) and bin 3 (thirty-one to sixty days) is where the cumulative exposure starts to correspond to a meaningful share of a single Shanghai summer; below that threshold, a buyer who tours the property may experience at most a handful of hot days and is plausibly indifferent. Second, the attenuation across bins 4 to 6 is what the dilution mechanism of Section 7 predicts rather than a counter-example to the cumulative-capitalization story: windows with one hundred fifty or more P90 days mechanically also stretch across two or more summers and large stretches of cool weather, so the average effect per cumulative day in those windows is mixed with non-summer observations that should have no penalty. The bins trace a non-linear cumulative effect that the linear baseline only approximates as a weighted average. On the cleanest subsamples (Section 7), the coefficient comes back close to the bin-3 magnitude.

5.3 Event-based indicators

We supplement the count measure with three binary indicators of heat events.

Indicator A (any extreme-heat day). Equal to one if the listing window contains at least one extreme-heat day.

Indicator B (heatwave). Equal to one if the listing window contains at least three extreme-heat days in total. We also tried the three-consecutive-day heatwave definition of [Kang et al. \(2024\)](#), attributed to the China Meteorological Administration; the interaction is likewise small and insignificant.



Notes: The horizontal axis is the midpoint of the cumulative extreme-heat bin in days. The vertical axis is the implied south-facing premium $\beta_1 + \theta_k$ in log points (zero-day bin as the reference). The shaded band is the 95 percent confidence interval. Specification follows equation (5).

FIGURE 2 South-facing premium across cumulative-exposure bins

Indicator C (high frequency). Equal to one if the share of extreme-heat days in the listing window exceeds the sample median.

Table 4 reports the south-facing-by-indicator interaction for each version, plus a joint specification with the cumulative-day measure and indicator B together. Indicator A (any extreme-heat day) and indicator C (high-frequency heat) give a negative and significant interaction (-0.0129 and -0.0048 , both $p < 0.01$). The three-consecutive-day heatwave indicator B is small and not significant (-0.0009). In the joint specification only the cumulative-day measure survives (-0.0059 , $p < 0.01$); the heatwave indicator is absorbed. What gets priced in is the running count of hot days over the window, not whether a single dramatic heatwave occurred: once the cumulative count is held fixed, the heatwave indicator adds nothing.

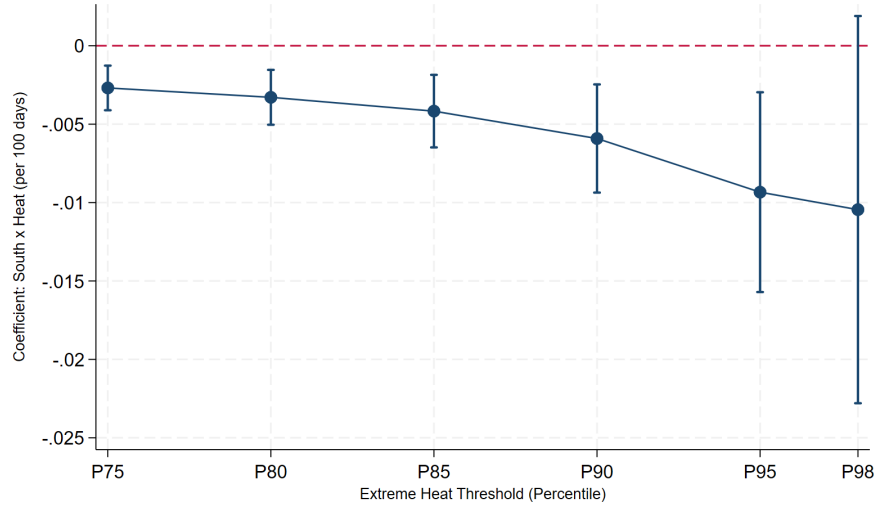
5.4 Dose-response across thresholds and degree-day measures

To check that the effect is not specific to the P90 cutoff, we re-estimate equation (2) with the heat measure computed at P75, P80, P85, P90, P95, and P98 thresholds. Figure 3 plots the resulting interaction coefficients with 95 percent confidence bands. The estimate slides smoothly from -0.0027 at P75 to -0.0033 (P80), -0.0042 (P85), -0.0059 (P90), -0.0093 (P95), and -0.0105 at P98; the confidence band widens at the tail because hot days at the higher cutoffs are sparser. The monotone pattern is consistent with the temperature-range responses in Deschênes and Greenstone (2011) and Kang et al. (2024).

TABLE 4 Event-based indicators

	(1) Any heat	(2) Heatwave	(3) High freq.	(4) Joint
South-facing (β_1)	0.02290*** (0.00221)	0.01207*** (0.00115)	0.01502*** (0.00167)	0.01615*** (0.00173)
Heat indicator (β_2)	0.00187 (0.00238)	0.00040 (0.00383)	-0.00032 (0.00212)	
South $\times$ Heat indicator (β_3)	-0.0129*** (0.0022)	-0.0009 (0.0039)	-0.0048*** (0.0017)	
Cumulative heat per 100 days (β_2^H)				-0.01332*** (0.00368)
South $\times$ Heat per 100 (β_3^H)				-0.0059*** (0.0018)
Heatwave B				0.00058 (0.00382)
South $\times$ Heatwave B				-0.0010 (0.0038)
Building controls	Yes (30)	Yes (30)	Yes (30)	Yes (30)
FEs	Yes	Yes	Yes	Yes
Observations	217,640	217,640	217,640	217,640

Notes: Columns 1 to 3 replace the cumulative heat measure with a binary indicator: any heat day (A), at least three extreme-heat days in total (B), or share of heat days above the sample median (C). Column 4 enters the cumulative measure and indicator B jointly. Standard errors two-way clustered by community and transaction date. Fixed effects contain community fixed effect, quarter fixed effect and duration fixed effect in all columns. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.



Notes: Each point is the south-facing-by-heat interaction estimated with the heat measure constructed at the indicated percentile of the daily mean temperature distribution. Bars are 95 percent confidence intervals.

FIGURE 3 $\hat{\beta}_3$ across heat-day percentile thresholds

A complementary check replaces the threshold-day count with cooling degree days (CDD) and heating degree days (HDD) accumulated over the listing window. For base temperature b , define

$$\text{CDD}_{i,t}^b = \frac{1}{100} \sum_{d \in [l_i, t_i]} \max(T_d^{\text{mean}} - b, 0), \quad \text{HDD}_{i,t}^b = \frac{1}{100} \sum_{d \in [l_i, t_i]} \max(b - T_d^{\text{mean}}, 0). \quad (6)$$

CDD accumulates degree-days only on days warmer than b , and HDD only on days cooler than b , so each metric scales with how far the daily mean reaches beyond the base rather than counting days above or below a fixed cutoff. This is the metric the energy literature uses for residential cooling and heating loads. We then re-estimate equation (2) replacing $\text{Heat}_{i,t}$ with $\text{CDD}_{i,t}^b$ (or with $\text{HDD}_{i,t}^b$) at four base values each. Table 5 reports the interaction of south-facing with CDD per 100 degree-days at four base temperatures (22, 24, 26, 28°C) and with HDD at four base temperatures (14, 16, 18, 20°C). The CDD interaction is negative at all four bases and grows in magnitude as the base rises (from -0.00037 at base 22°C to -0.00194 at base 28°C), the same monotone shape we obtained with the percentile count. The HDD interaction is small and weakly negative (-0.00012 to -0.00025), about an order of magnitude smaller than the CDD coefficients. Cold weather during the listing window does little to the premium, while heat does most of the work. This fits a setting where summer heat, not winter cold, is what moves the premium at Shanghai’s latitude.

5.5 Apparent temperature

The baseline measures heat with the dry-bulb daily mean. Apparent temperature folds humidity and wind into that mean and comes closer to what a buyer’s body registers during a visit. We reconstruct apparent temperature from the station record using the [Steadman \(1994\)](#) formula in its non-radiation form,

$$\text{AT}_d = T_d^{\text{mean}} + 0.33 e_d - 0.70 w_d - 4.00, \quad (7)$$

where T_d^{mean} is the daily mean temperature, w_d the daily mean wind speed, and e_d the water-vapour pressure in hPa, recovered from relative humidity through the Magnus relation $e_d = (\text{RH}_d/100) \cdot 6.105 \cdot \exp(17.27 T_d^{\text{mean}}/(237.7 + T_d^{\text{mean}}))$. Apparent temperature tracks the dry-bulb mean closely over the full year (correlation 0.994) but pulls apart in summer, when humidity is high and wind is low: the June-to-August mean is 31.0°C in apparent terms against 27.5°C dry-bulb. We then rebuild the cumulative heat measure on apparent temperature, using both the degree-day construction of equation (6) (AT-CDD at base 26°C,

TABLE 5 CDD and HDD: full coefficients across base temperatures

	Base 1	Base 2	Base 3	Base 4
<i>Panel A. Cooling degree days (CDD), base = 22, 24, 26, 28°C</i>				
South (β_1)	0.01640*** (0.00169)	0.01635*** (0.00170)	0.01615*** (0.00172)	0.01560*** (0.00175)
CDD per 100 d-d (β_2)	-0.00139*** (0.00019)	-0.00196*** (0.00032)	-0.00269*** (0.00061)	-0.00222 (0.00141)
South $\times$ CDD (β_3)	-0.00037*** (0.00010)	-0.00058*** (0.00016)	-0.00099*** (0.00029)	-0.00194*** (0.00066)
<i>Panel B. Heating degree days (HDD), base = 14, 16, 18, 20°C</i>				
South (β_1)	0.01434*** (0.00164)	0.01441*** (0.00165)	0.01451*** (0.00165)	0.01462*** (0.00166)
HDD per 100 d-d (β_2)	-0.00046** (0.00021)	-0.00038** (0.00015)	-0.00034*** (0.00011)	-0.00031*** (0.00009)
South $\times$ HDD (β_3)	-0.00025** (0.00012)	-0.00019** (0.00009)	-0.00015** (0.00007)	-0.00012** (0.00005)
Building controls	Yes (30)	Yes (30)	Yes (30)	Yes (30)
FEs	Yes	Yes	Yes	Yes
Observations	217,640	217,640	217,640	217,640

Notes: Each column is a separate regression. Panel A replaces the cumulative heat days in equation (2) with CDD per 100 degree-days at the indicated base temperature (22, 24, 26, 28°C in columns 1 to 4). Panel B does the same for HDD at base 14, 16, 18, 20°C. Other covariates and fixed effects match the baseline. Standard errors are two-way clustered by community and transaction date. Fixed effects contain community fixed effect, quarter fixed effect and duration fixed effect in all columns. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

AT-HDD at base 18°C) and a P90 extreme-day count that mirrors the baseline.

Table 6 reports the three interactions. The south-facing-by-AT-CDD interaction is negative and significant, as is the south-facing-by-extreme-AT-heat interaction; the latter, scaled to per-hundred-day units, is -0.0058 , almost identical to the dry-bulb baseline of -0.00592 . The AT-HDD interaction is an order of magnitude smaller, consistent with the dry-bulb degree-day pattern in Table 5. The penalty is the same whether heat is measured by the dry-bulb mean or by an index that already folds in humidity and wind. Were the result an artifact of any one weather variable, the two measures would not line up this closely.

TABLE 6 Robustness: apparent temperature (Steadman 1994)

	(1) AT-CDD (base 26)	(2) AT-HDD (base 18)	(3) Extreme AT hot (P90)
South (β_1)	0.01657*** (0.00172)	0.01454*** (0.00165)	0.01640*** (0.00174)
AT measure (β_2)	-0.0000158*** (0.0000023)	-2.53e-06*** (8.16e-07)	-0.0001844*** (0.0000363)
South $\times$ AT measure	-4.05e-06*** (1.10e-06)	-1.07e-06** (4.80e-07)	-0.0000580*** (0.0000164)
Building controls	Yes (30)	Yes (30)	Yes (30)
FEs	Yes	Yes	Yes
Observations	217,640	217,640	217,640

Notes: Apparent temperature is constructed from equation (7). Columns 1 and 2 use AT-based cooling and heating degree days (per day, base 26°C and 18°C). Column 3 counts days above the 90th percentile of apparent temperature within the listing window. Coefficients are in per-day units; the extreme-AT-heat interaction in column 3 corresponds to -0.0058 per hundred days, close to the dry-bulb baseline of -0.00592. Standard errors two-way clustered by community and transaction date. Fixed effects contain community fixed effect, quarter fixed effect and duration fixed effect in all columns. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6 Mechanism

6.1 Thermal discomfort

What buyers experience during property visits is not air temperature alone but the perceived heat that combines temperature with humidity, wind, and sky cover. The physiology behind this combined sensation is well established in the indoor-climate literature: [Chen et al. \(2015\)](#) and [Lin and Deng \(2008\)](#) show that humidity, air movement, and radiative load enter the comfort equation alongside dry-bulb temperature. We add interactions of the cumulative heat measure and the south-facing indicator with three modifiers chosen on physical grounds: humidity (the body cools less efficiently in humid air), wind (convective cooling reduces indoor and outdoor heat load), and cloud cover (which reduces direct solar radiation through south-facing windows).

The specification expands equation (2) into a triple-interaction form,

$$\begin{aligned}
\ln(P_{i,t}) = & \alpha + \beta_1 \text{South}_i + \beta_2 \text{Heat}_{i,t} + \beta_3 \text{South}_i \times \text{Heat}_{i,t} \\
& + \gamma_1 M_{i,t} + \gamma_2 \text{South}_i \times M_{i,t} + \gamma_3 \text{Heat}_{i,t} \times M_{i,t} \\
& + \gamma_4 \text{South}_i \times \text{Heat}_{i,t} \times M_{i,t} + X'_i \gamma + \delta + \varepsilon_{i,t},
\end{aligned} \tag{8}$$

where $M_{i,t}$ is the modifier. $\gamma_4 < 0$ means the modifier amplifies the heat penalty on south-facing units; $\gamma_4 > 0$ means the modifier offsets it.

Table 7 reports the four specifications. Columns 1 and 2 use the listing-window count of high-wind and low-sunshine (cloudy) days; columns 3 and 4 use the listing-window mean level of humidity and wind speed, centered at the sample mean. The pattern is consistent with thermal-comfort logic. Humidity amplifies the south-facing-by-heat interaction ($\hat{\gamma}_4 = -0.0025$, $p < 0.01$). Wind offsets it ($\hat{\gamma}_4 = +0.0597$, $p < 0.01$). Cloudiness offsets it ($\hat{\gamma}_4 = +0.0184$, $p < 0.01$). An additional specification for extreme rainfall (which usually coincides with cooler air) gives $\hat{\gamma}_4 = +0.0193$, $p < 0.01$.

TABLE 7 Mechanism: humidity, wind, and cloud cover as thermal-comfort modifiers

	(1) Windy (count)	(2) Cloudy (count)	(3) Humidity (continuous)	(4) Wind speed (continuous)
South (β_1)	0.02429*** (0.00202)	0.02432*** (0.00205)	0.01696*** (0.00185)	0.01668*** (0.00178)
Heat per 100 days (β_2)	0.04427*** (0.00881)	0.06809*** (0.00828)	-0.01682*** (0.00513)	-0.02528*** (0.00560)
South $\times$ Heat (β_3)	-0.0190*** (0.00396)	-0.0237*** (0.00405)	-0.0085*** (0.00219)	-0.0090*** (0.00215)
South $\times$ Heat $\times$ M (γ_4)	+0.0306*** (0.00443)	+0.0184*** (0.00271)	-0.0025*** (0.00093)	+0.0597*** (0.02111)
Direction of γ_4	offsets	offsets	amplifies	offsets
Building controls	Yes (30)	Yes (30)	Yes (30)	Yes (30)
FEs	Yes	Yes	Yes	Yes
Observations	217,640	217,640	217,640	217,640

Notes: Each column estimates equation (8) with the modifier M shown in the header. The dependent variable is the log unit transaction price (sample mean 1.49). Columns 1 and 2 measure wind and cloud cover by the listing-window count of high-wind and low-sunshine days; columns 3 and 4 measure humidity and wind speed by the listing-window mean level, centered at the sample mean. The coefficient of interest is the triple interaction γ_4 : a negative sign means the modifier amplifies the south-facing heat penalty, a positive sign means it offsets the penalty. Humidity amplifies the penalty, while wind and cloud cover offset it, the pattern thermal-comfort physics predicts. Building controls, community, year-by-quarter, and listing-duration fixed effects are included throughout. Standard errors in parentheses, two-way clustered by community and transaction date. Fixed effects contain community fixed effect, quarter fixed effect and duration fixed effect in all columns. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The continuous specifications in columns 3 and 4 are the cleanest because the modifier varies smoothly and is uncorrelated with the heat-day count by construction (we centered it at the mean). The sign on humidity is the opposite of the sign on wind, which is what the physics would predict: more humid air weakens the body’s ability to lose heat through evaporation, while more wind speeds up convective cooling near windows.

The three modifiers correspond to three physical mechanisms that the thermal-comfort literature has identified as the most important moderators of perceived heat besides dry-bulb temperature itself (Chen et al., 2015; Lin and Deng, 2008). Humidity controls how efficiently the body can dissipate heat through sweat evaporation. At Shanghai’s relative humidity in summer (typically 75 to 85 percent at peak), the sweat-evaporation channel is already partially blocked, and any additional humidity makes the same air temperature feel hotter. Wind drives forced convection at the building envelope and around the body; a south-facing room with the windows open in a breeze cools much faster than the same room in still air. Cloud cover blocks direct solar radiation, the channel that south-facing windows differentially admit; a cloudy summer day is one in which the orientation gap in indoor temperature largely closes. The three signs we recover for humidity, wind, and cloud cover line up with all three of these physical predictions. The offsetting effect of wind ($\gamma_4 = +0.0597$) is large enough that the penalty shrinks toward zero in the windiest listing windows. A thermal-comfort reading expects exactly this: moving air carries heat away from the building envelope, so the gap between orientations narrows when wind is high.

A further check uses extreme-rainfall days as a modifier. Heavy-rain days in Shanghai are systematically cooler than the seasonal average because they are associated with cloud cover and the latent-heat sink of precipitation. The triple interaction of south-facing, cumulative heat, and the count of extreme-rainfall days within the window gives $\gamma_4 = +0.0193$ ($p < 0.01$), the same sign as the wind and cloud modifiers and roughly the same magnitude as the cloudy-day result. This is a useful consistency check because rain is generated by atmospheric dynamics that are largely orthogonal to the building-attribute side of the regression. Four modifiers driven by different atmospheric processes all move the penalty in the direction their effect on indoor heat would imply. A single omitted variable standing in for the whole result would have to reproduce all four of these patterns at once, which is hard to arrange.

A single-visit story, in which buyers inspect on the hottest day and update once, predicts β_3 to track visit-day weather rather than listing-period averages, and would not predict the three modifier signs to line up with comfort physics. A general-demand story, in which hot summers depress all prices, is absorbed by the community-quarter contrast that identifies β_3 from the within-cell south-versus-non-south differential.

6.2 Expected versus realized heat

The cumulative heat measure mixes two parts. One is the heat that is predictable from the calendar months the listing window covers, which a buyer can forecast from a normal year. The other is the heat that actually occurred during the window, measured as the

realized count minus that forecast. If buyers price the discomfort of a normal summer that the calendar implies, the predictable part should carry the effect. If only the heat they happened to live through matters, the realized part should dominate. A joint specification separates the two.

To separate them, we decompose the cumulative heat measure into an expected component and a residual component. The expected component for transaction i is the climatological probability of an extreme-heat day in the calendar months covered by the listing window, multiplied by the window length:

$$\text{Heat}_{i,t}^{\text{exp}} = \bar{\rho}_{m(l_i,t_i)} \cdot D_{i,t}, \quad (9)$$

where $\bar{\rho}_m$ is the long-run average daily extreme-heat probability in calendar month m over the 2000 to 2020 station record, $m(l_i, t_i)$ is the set of months covered by the listing window, and $D_{i,t}$ is the window length in days. The residual is $\text{Heat}_{i,t}^{\text{res}} = \text{Heat}_{i,t} - \text{Heat}_{i,t}^{\text{exp}}$. We enter both measures, together with their interactions with the south-facing indicator, into the baseline regression.

TABLE 8 Expected versus realized heat

	Coefficient	Std. error	t
South $\times$ Expected heat (β_3^{exp})	−0.00551***	0.00172	−3.20
South $\times$ Realized heat (β_3^{res})	−0.00696***	0.00191	−3.64
Wald test of equality $\beta_3^{\text{exp}} = \beta_3^{\text{res}}$:	$F(1, 2836) = 2.73$ ($p = 0.099$)		
Building controls	Yes (30)	Yes (30)	Yes (30)
FEs	Yes	Yes	Yes
Observations	217,640	217,640	217,640

Notes: The cumulative heat measure is split into an expected component (equation (9)), equal to the climatological extreme-heat probability of the calendar months the window spans times the window length, and a residual component equal to realized minus expected heat. Each is interacted with the south-facing indicator and the two are entered jointly in the baseline regression. The table reports the two interactions of interest; the south indicator, both heat main effects, the full building controls, and the community, year-by-quarter, and listing-duration fixed effects are included but not shown. The Wald test does not reject equality of the two interaction coefficients at conventional levels, so the predictable and realized parts carry similar weight per day. Standard errors two-way clustered by community and transaction date. Fixed effects contain community fixed effect, quarter fixed effect and duration fixed effect in all columns. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8 reports the two interactions. Both interactions are negative and significant.

The predictable part gives -0.00551 (clustered SE 0.00172 , $t = -3.20$) and the realized part gives -0.00696 (SE 0.00191 , $t = -3.64$). A Wald test of equality $\mathcal{H}_0 : \beta_3^{\text{exp}} = \beta_3^{\text{res}}$ gives $F(1, 2836) = 2.73$ with $p = 0.099$, so the two cannot be told apart at conventional levels. The two components carry similar weight per day, and the Wald test cannot separate them. Neither sits easily with a frictionless benchmark, under which the value of a durable orientation depends on the climate the owner faces after purchase and not on the weather that fell inside one listing window. The realized part makes this sharpest: it is the weather surprise relative to the seasonal norm, so it carries no information about the future climate, and the one rational reading of it, that buyers learn an unobserved thermal trait from the hot days they live through, would load on the transaction price rather than the listing price, which Section 6.3 does not support. We therefore read the response as carrying an extrapolative element, with the premium marked down as recent heat accumulates, rather than as capitalization of a fixed climate alone. The predictable component is identified only from the calendar months the window spans and its length, not from realized weather, so it is the cleaner of the two for identification: that it carries a negative interaction even when entered jointly with the realized term shows the result does not rest on realized-weather variation that might correlate with time on the market.

6.3 Listing price and discount rate

The baseline uses the transaction price as the dependent variable. To pin down which side of the transaction prices in the heat penalty, we re-estimate with the log listing price and with the discount rate as the outcome. The listing price is the seller’s posted price at the start of the window, set before any of the cumulative heat is realized for windows that begin in spring or early summer, and updated through subsequent revisions for windows that drag on. The discount rate is the proportional reduction between listing and transaction prices and is therefore a measure of bargaining at the close. If the heat penalty were a buyer-side effect, sentiment shocks on the day of signing or a buyer’s negotiation power growing with the heat experienced during visits, we would expect the discount-rate coefficient to load and the listing-price coefficient to be small. If the penalty were sellers pricing in the heat cost early, it would appear in the listing price and the discount rate would not respond.

To compare these magnitudes on a consistent footing, we estimate all three outcomes on a common sample, trimmed identically to drop the small share of records with an absolute discount above one half ($N = 212,119$), so that the coefficients satisfy the identity $\beta_3^{\text{disc}} \approx \beta_3^{\ln L} - \beta_3^{\ln T}$. The discount rate, which compares the posted and final price of the same transaction, does not move with heat (-0.0005 , $t = -0.72$). Consistent with that tight zero,

TABLE 9 Pricing margin: listing price, transaction price, and discount rate

Dependent variable	South (β_1)	Heat (β_2)	South $\times$ Heat (β_3)
ln unit listing price	0.01582*** (0.00181)	-0.01353*** (0.00369)	-0.00781*** (0.00180)
ln unit transaction price	0.01640*** (0.00170)	-0.01365*** (0.00362)	-0.00705*** (0.00167)
Discount rate	-0.00036 (0.00066)	0.00029 (0.00097)	-0.00051 (0.00070)
Building controls	Yes (30)	Yes (30)	Yes (30)
FEs	Yes	Yes	Yes
Observations	212,119	212,119	212,119

Notes: All three outcomes are estimated on a common sample, trimmed identically to $|\text{discount}| < 0.5$ ($N = 212,119$); records with a listing-to-transaction gap above one half are dropped as likely data-entry errors, and the common trim lets the three regressions be compared on the same observations, so that $\beta_3^{\text{disc}} \approx \beta_3^{\text{ln } L} - \beta_3^{\text{ln } T}$ holds across rows. The listing-minus-transaction gap in β_3 is -0.0008 ($t = -0.90$, $p = 0.37$), statistically zero and consistent with the discount-rate coefficient. The full-sample transaction baseline (without the discount trim) is -0.00592 (Table 3). Standard errors two-way clustered by community and transaction date. Fixed effects contain community fixed effect, quarter fixed effect and duration fixed effect in all columns. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

the listing-price and transaction-price interactions are nearly the same size (-0.0078 , $t = -4.33$, and -0.0070 , $t = -4.22$, each significant at the one percent level), and their difference is -0.0008 ($t = -0.90$, $p = 0.37$), indistinguishable from zero; a stacked specification that lets every fixed effect and control enter separately for each equation returns the same gap. The penalty is set in the seller’s listing price and carries through to the close at the same magnitude, with no further adjustment in the discount. This fits the broader evidence on the centrality of seller behavior in housing price formation (Genesove and Mayer, 2001).

The presence of the penalty in the listing price also rules out a class of alternative explanations. Consider a story in which the heat penalty reflects only short-run buyer fatigue at the moment of signing, as in the day-of-deal sentiment results of Hu and Lee (2020) or the polluted-day evidence of Qin et al. (2019). Under that mechanism the listing price, posted weeks or months before the deal, should be largely unaffected by deal-day weather, and the penalty should show up only in the discount rate. We find neither pattern: the listing price already carries the penalty and the discount rate does not move. A seller who priced only the long-run climate a buyer faces after purchase would not respond to the particular heat realized while the unit sits on the market, so the pattern is most naturally read as extrapolative pricing by sellers who over-weight recently experienced heat. We cannot fully rule out a rational-learning channel on the buyer side, but the flat discount rate gives it little

room: heat does not widen the gap between the posted and final price. A related question is whether weather close to the signing date, rather than over the whole window, is what moves the price. We split the listing-window heat into an early portion and a portion falling in the last one to four weeks before signing, and enter both interacted with south-facing. In the transaction price, the early portion carries a negative south interaction while the deal-window portion carries a positive one. In the listing price the deal-window interaction is essentially zero, and only the early portion remains negative. The listing price is posted before the deal-window weather is realized, so a valuation effect should not appear there, and it does not: what enters the listing price is the cumulative early-window heat, which is the object behind our main estimate. The positive deal-window term in the transaction price does not pass through to the listing price and shows up instead in the closing discount, where heat near the signing date raises the discount on all units regardless of orientation. We read this as a closing-timing effect in bargaining rather than a part of how orientation is valued, and we do not treat it as evidence on the orientation premium itself.

7 Robustness

7.1 Listing-period dilution

The cumulative measure adds up extreme-heat days within the entire listing window. When that window stretches across several seasons or several years, the count blurs together the hot and cool months. The baseline estimate is therefore a dilution-weighted average of the true effect inside a single hot season and zero outside it. To check this, we cut the sample along four dilution measures: the number of summers in the listing window, the listing duration in days, the number of calendar years spanned, and the intersection of the cleanest of these conditions.

Table 10 reports $\hat{\beta}_3$ in each subsample. The magnitudes scale up by a factor of nine to twelve in the single-summer or short-listing subsamples, and shrink to near zero in the multi-year, long-listing tail. The cleanest sample, the one with a single calendar year and a listing duration between thirty-one and one hundred and eighty days, gives $\hat{\beta}_3 = -0.0618$, about ten times the baseline.

The dilution pattern matters for how readers should compare our coefficient to those in the broader climate-capitalization literature. Studies that use city-year level data, such as Kang et al. (2024), compress all summer exposure of a city into a single observation per year and lose the within-city variation in actual unit-level experience. Studies that use a fixed deal-window of, say, fifteen days, such as Chen et al. (2023), restrict attention to a

tight slice around the transaction that necessarily falls in a single season. Both designs limit dilution by construction, the first by averaging it out and the second by sampling it away. Our single-summer subsample applies the equivalent correction inside our longer window: by restricting to listings that span only one summer, the cumulative count of P90 days within the window is no longer averaged against cool months from a second or third season, and the coefficient rises by roughly an order of magnitude. Coefficients from different climate-capitalization designs are not directly comparable without accounting for how each design measures exposure. The size of our estimate depends on how much of the listing window falls inside a single summer, the same measurement issue that separates a city-year design from a short deal-window design. Because the single-summer subsamples also select on time-on-market, this magnitude combines reduced dilution with faster-selling listings. We therefore treat the full-sample -0.0059 as the baseline estimate and the single-summer figure as an upper bound.

7.2 Heat versus listing duration

South-facing units sell faster, so their listing windows are shorter and accumulate fewer extreme-heat days. The heat measure is therefore correlated with orientation through duration, and a reader may worry that the interaction picks up a duration effect rather than an exposure effect. We probe this in several ways and report all of them, including the ones that cut against a clean separation.

The interaction survives every control for duration that does not force heat and duration to compete head to head. Refining the duration fixed effect from five bands to ten, twenty, and forty leaves the coefficient between -0.0060 and -0.0064 (all significant at one percent), and a cubic polynomial in duration entered directly gives -0.0083 . Replacing the count with the listing-period heat share, which is duration-free by construction, gives -0.0207 . Within the single duration band from sixty-one to one hundred and eighty days the coefficient is -0.050 , and the middle bands of a five-way split run from -0.05 to -0.06 . Across these specifications the penalty is present and, if anything, larger than the baseline once dilution is reduced.

Two specifications show the limit of what we can claim. When we add the interaction of south-facing with duration and its square, so that heat and duration compete directly, the heat interaction turns positive ($+0.0137$). When we instead residualize heat on a flexible function of duration and use only the orthogonal part, the coefficient keeps its sign but loses significance (-0.0063 , $z = -1.17$). Heat and duration are collinear enough in the interaction that a direct decomposition cannot assign the effect to one or the other. We therefore do not

claim to have separated heat from duration cleanly. What we can say is that the penalty does not disappear under finer duration fixed effects, under a duration-free intensity measure, or within narrow duration bands, which is the pattern a duration artifact would not produce.

TABLE 10 Listing-period dilution: full coefficients across subsamples

Subsample	South (β_1)	Heat (β_2)	South $\times$ Heat (β_3)	Obs.
Baseline (full sample)	0.01610*** (0.00173)	-0.01333*** (0.00368)	-0.00592*** (0.00176)	217,640
<i>Panel A. Number of summers spanned</i>				
One summer	0.02698*** (0.00336)	0.04590*** (0.00962)	-0.0537*** (0.00872)	83,381
Two summers	0.02272** (0.00931)	0.18724*** (0.01796)	-0.0181 (0.01111)	41,719
Three or more summers	0.01943*** (0.00706)	-0.01292** (0.00544)	-0.0012 (0.00478)	33,010
<i>Panel B. Listing duration</i>				
≤ 90 days	0.02525*** (0.00259)	0.01652 (0.02007)	-0.0373* (0.02006)	67,220
≤ 180 days	0.02432*** (0.00229)	0.03267*** (0.01188)	-0.0643*** (0.01125)	93,456
> 365 days	0.01013*** (0.00323)	-0.00647 (0.00483)	+0.0018 (0.00291)	84,392
<i>Panel C. Number of calendar years spanned</i>				
One year	0.02801*** (0.00239)	0.04520*** (0.00884)	-0.0724*** (0.00781)	93,501
Two years	0.01645*** (0.00314)	0.08892*** (0.01268)	-0.0181*** (0.00493)	63,232
Three or more years	0.00844** (0.00406)	-0.02023*** (0.00379)	+0.0047 (0.00322)	56,727
<i>Panel D. Clean subsamples (one year and listing in window)</i>				
≤ 90 days	0.02601*** (0.00263)	0.01880 (0.02007)	-0.0399** (0.02001)	63,641
[31, 180] days	0.02483*** (0.00291)	0.04293*** (0.01317)	-0.0618*** (0.01258)	45,611

Notes: Each row re-estimates equation (2) in the indicated subsample. Building controls (30), community FE, year-by-quarter FE, and listing-duration-group FE included throughout. Standard errors (in parentheses) two-way clustered by community and transaction date. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7.3 Building height tiers

If the mechanism is physical cooling, the heat penalty should be larger in buildings that lack modern insulation. Older walk-up buildings in Shanghai tend to have six or fewer floors, single-pane windows, and no central air conditioning. Tall elevator buildings tend to have better insulation. We sort buildings by total number of floors into three tiers: 6 or fewer floors, 7 to 18 floors, and more than 18 floors. We then estimate the interaction between south-facing, heat, and tier dummies and compute the implied $\hat{\beta}_3$ in each tier with a linear combination.

TABLE 11 Building height: the heat penalty by building tier

	≤ 6 floors (low-rise)	7 to 18 floors	> 18 floors (high-rise)
<i>Panel A. Heat penalty within tier (implied β_3)</i>			
South $\times$ Heat (β_3)	−0.0149*** (0.0028)	−0.0079** (0.0034)	−0.0013 (0.0028)
<i>Panel B. Difference from the low-rise tier</i>			
Tier $\times$ South $\times$ Heat	(ref.)	+0.0070 (0.0045)	+0.0136*** (0.0039)
Building controls	Yes (30)	Yes (30)	Yes (30)
FEs	Yes	Yes	Yes
Observations	217,640	217,640	217,640

Notes: The table comes from one regression that interacts the south-facing indicator and cumulative heat with two building-height tier dummies, with the low-rise tier (six floors or fewer) as the reference; tiers are read from the total-floor field on listings. Panel A reports the south-by-heat penalty for each tier, recovered as a linear combination of the base interaction and the relevant tier interaction (Stata `lincom`); this is the quantity discussed in the text. Panel B reports the triple-interaction term, that is, the difference in the penalty between each tier and the low-rise reference. The penalty falls from −0.0149 in low-rise stock to a value indistinguishable from zero above eighteen floors, and the gap between the tall tier and the low-rise tier is significant at one percent. The regression also includes the tier main effects, the tier-by-south terms, and the tier-by-heat terms, which are estimated but not shown. The south premium at zero heat is 0.0204 (SE 0.0026) in the low-rise tier and is similar in size across tiers. A continuous version that interacts south-facing and heat with the centered total floor count gives a triple-interaction coefficient of +0.00048 ($p < 0.01$), the same gradient in continuous form. Standard errors two-way clustered by community and transaction date. Fixed effects contain community fixed effect, quarter fixed effect and duration fixed effect in all columns.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The heat penalty declines monotonically from low-rise to high-rise. The estimate in the

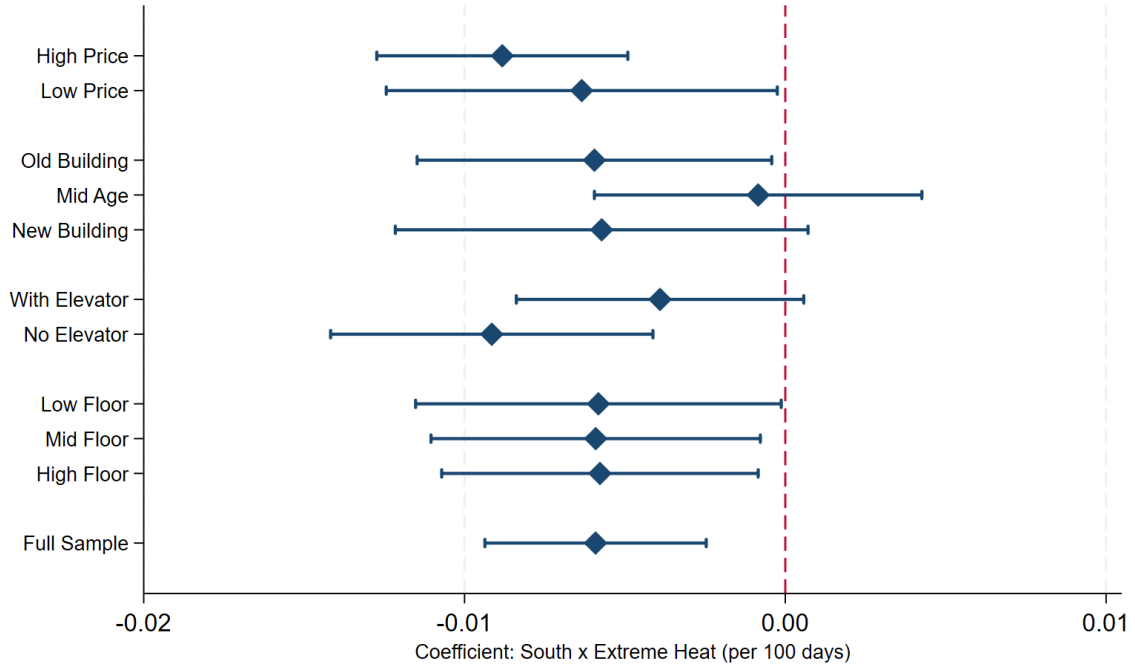
low-rise tier is about 2.5 times the full-sample baseline. The estimate in the tall-building tier is close to zero. The pattern is the one a physical-cooling mechanism would predict. Low-rise stock in Shanghai is largely composed of pre-1990s walk-up apartments with single-pane windows and weak insulation, while tall buildings built since the 2000s usually have double-pane windows, structural shading, and central or split air-conditioning systems. The same one hundred days of P90 exposure translates into very different indoor heat loads across these two building types, and the heat penalty on the south-facing orientation tracks that gap.

Figure 4 extends the same exercise across other cuts of the sample (price quartile, building age band, elevator status, floor band, full sample). The point estimates cluster around the baseline with one informative exception: no-elevator units, which proxy for older walk-up stock, give a coefficient about 1.5 times the full-sample value, while elevator units give a coefficient about a third smaller and statistically indistinguishable from zero. The elevator margin reinforces the height-tier result. Buildings with elevators in China are overwhelmingly taller and newer; the elevator dummy therefore captures much of the same insulation-and-shading variation that drives the height-tier pattern. The fact that the heat penalty is concentrated in no-elevator units, even after controlling for price quartile, building age, and floor band, points squarely at the physical thermal channel rather than at a buyer-type or neighborhood-quality story.

7.4 Orientation placebo

The story requires that the heat penalty operates through direct solar radiation into south-facing rooms. North-facing rooms do not receive direct sunlight at noon in Shanghai, so they should not show a comparable interaction. East-facing and west-facing rooms receive direct morning and afternoon sunlight respectively, and should show a smaller version of the effect. We replace the south-facing indicator in the baseline with east-facing, west-facing, and north-facing indicators (with all four orientations entered jointly). This is a placebo in the sense that it asks whether a generic explanation (such as buyer fatigue from many summer visits, or a seasonal listing pattern correlated with heat) would also produce a negative interaction on the non-south orientations. Under any such generic story, all orientations should respond similarly to listing-period heat; under the direct-solar-radiation story, the sign should differ across orientations and align with the time of day each orientation receives direct sun.

The physics is asymmetric across orientations. South, east, and west all receive direct solar radiation at some part of a Shanghai summer day, so each should carry a negative heat



Notes: Each row is the south-by-heat coefficient estimated on the indicated subsample, with 95 percent confidence intervals. The bottom row is the full-sample baseline.

FIGURE 4 $\hat{\beta}_3$ across sample cuts

interaction. North does not, so it should not. Table 12 delivers exactly that: south, east, and west are negative (-0.00378 , -0.00947 , -0.00705), and north is positive ($+0.01185$). Buyer fatigue or a seasonal listing pattern would not tell the orientations apart and would push all four coefficients the same way. North is the one orientation that gets no direct summer sun at midday, and it is the one orientation whose premium does not fall. That single sign flip, significant at the ten percent level, is the sharpest qualitative evidence that the channel is direct solar radiation.

7.5 Listing-season heterogeneity

If the mechanism is buyer experience of heat during the listing window, the timing of the listing within the year should matter. A listing posted in summer captures heat exposure from the start; a listing posted in spring or autumn accumulates heat over its first summer in the market; a listing posted in winter that transacts within a few months may avoid summer altogether. We split the sample by the season in which the property was first listed and re-estimate the baseline.

The interaction is largest in magnitude for listings posted in summer (-0.0117), where the

TABLE 12 Orientation placebo: four orientations entered jointly

	Coefficient	Std. error
<i>Main effects:</i>		
South	0.00056	0.00190
East	−0.02886***	0.00326
West	−0.03310***	0.00346
North	−0.09693***	0.00686
Heat per 100 days	−0.01368***	0.00412
<i>Interactions with cumulative heat:</i>		
South × Heat	−0.00378*	0.00212
East × Heat	−0.00947***	0.00354
West × Heat	−0.00705*	0.00371
North × Heat	+0.01185*	0.00676
Building controls	Yes (30)	
FEs	Yes	
Observations	217,640	

Notes: All four orientation dummies and their interactions with heat enter the regression jointly. The omitted reference category is units without a south, east, west, or north designation; "South" follows the baseline definition (south or south-north). Specification otherwise matches the baseline. Fixed effects contain community fixed effect, quarter fixed effect and duration fixed effect in all columns.

TABLE 13 Listing season: full coefficients

	Baseline	Listed summer	Listed spring/autumn	Listed winter
South (β_1)	0.01610*** (0.00173)	0.01785*** (0.00407)	0.01578*** (0.00198)	0.01628*** (0.00346)
Heat per 100 days (β_2)	−0.01333*** (0.00368)	−0.00789 (0.00787)	0.00267 (0.00667)	−0.03983*** (0.00490)
South × Heat (β_3)	−0.00592*** (0.00176)	−0.01166** (0.00533)	−0.00956*** (0.00229)	−0.00102 (0.00288)
Building controls	Yes (30)	Yes (30)	Yes (30)	Yes (30)
FEs	Yes	Yes	Yes	Yes
Observations	217,640	39,761	133,695	39,759

Notes: Columns split by the season in which the property was first posted. Summer covers June–August, winter covers December–February, spring/autumn covers the remaining months. Standard errors two-way clustered. Fixed effects contain community fixed effect, quarter fixed effect and duration fixed effect in all columns. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

buyer experiences heat from day one, and remains significantly negative for spring/autumn listings (-0.0096) whose windows accumulate hot days during the next summer. Listings posted in winter give an interaction of -0.0010 that is statistically zero. Listings that never sit through a summer show no penalty; those that sit through one carry the full effect. When the heat lands matters, which is what we would expect if the penalty builds up over the window rather than landing on the day the contract is signed.

7.6 Dropping top and bottom floors

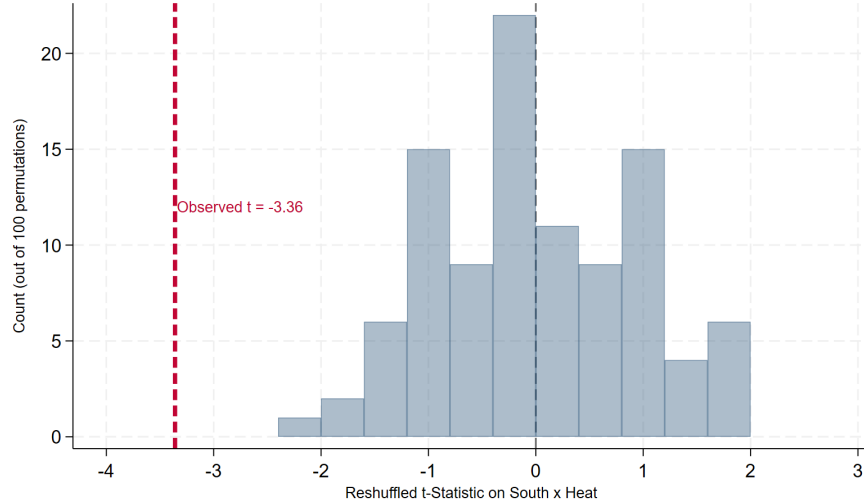
Top-floor units are more exposed to roof heat in summer; bottom-floor units may suffer from damp and shading that confound the orientation channel. To check that these extreme floors are not driving the result, we re-estimate the baseline (i) dropping units recorded as either top or bottom floor and (ii) keeping only middle-floor units. The interaction is -0.00661^{***} (SE 0.0022) after dropping top and bottom floors, and -0.00591^{**} (SE 0.0026) on middle-floor units only. Both sit within sampling error of the baseline (-0.00592^{***}), so the result is not an artifact of unusual floor positions.

7.7 Other robustness checks

A few shorter checks remain. Adding total floors of the unit’s building to the control vector leaves $\hat{\beta}_3$ unchanged at the sixth decimal place; the height-tier results in Table 11 are not driving the baseline. Replacing community fixed effects with business-zone fixed effects gives $\hat{\beta}_3 = -0.0130^{***}$, more negative than the baseline. Community fixed effects therefore absorb part of the variation we care about, and the baseline can be read as a lower bound. Refining the listing-duration-group fixed effect from five bands to ten, twenty, and forty bands gives $\hat{\beta}_3$ between -0.0060 and -0.0065 across all four specifications, which rules out a coarse-binning artifact in the duration control. Replacing the cumulative count with the listing-period heat share, defined as extreme-heat days divided by total listing days, gives $\hat{\beta}_3 = -0.0207^{**}$ with no duration control needed because the share is duration-independent. A randomization placebo re-estimates the interaction on data in which the heat measure is reshuffled. Let t^{obs} denote the observed statistic on β_3 and $\{t_b^{\text{fake}}\}$ the statistics from the reshuffled samples; the permutation p -value is

$$p^{\text{perm}} = \frac{1}{B} \sum_{b=1}^B \mathbf{1} \left(\left| t_b^{\text{fake}} \right| \geq \left| t^{\text{obs}} \right| \right). \quad (10)$$

Our primary version permutes the heat measure within community, holding the community structure fixed, with $B = 500$; none of the reshuffled samples produces an interaction as large as the observed one, so $p^{\text{perm}} = 0$ on both the coefficient and the t -statistic. A coarser version that reshuffles across all observations ($B = 100$) gives a fake- t distribution centered near zero with range $[-2.46, 2.86]$, against an observed $|t^{\text{obs}}| = 3.36$, again $p^{\text{perm}} = 0$ (Figure 5). The interaction is not an artifact of the heat measure’s distribution.



Notes: Histogram of t -statistics on the south-by-heat interaction from 100 across-observation reshuffles of the cumulative-heat measure (the coarser secondary check; the primary within-community test with $B = 500$ is reported in the text). The red dashed line marks the observed $t = -3.36$ from the baseline.

FIGURE 5 Permutation distribution of the south-by-heat t -statistic

Listing-date reliability. A share of early-window observations carry a listing date that coincides with the start of the data panel, a pattern consistent with a default value filled in when the original listing date was missing. These records have a mechanically long window and a high cumulative-heat count. Dropping the observations whose listing date falls on this default value gives $\hat{\beta}_3 = -0.0094$ (SE 0.0019), and dropping the broader set of observations that pile up at the panel start gives $\hat{\beta}_3 = -0.0126$ (SE 0.0021); both are larger in magnitude than the full-sample estimate and significant at the one percent level. The penalty is therefore not an artifact of these records. We retain them in the baseline for completeness and report the listing-window construction in full in the data appendix.

Selection. The selection concern is that south-facing units may delist during hot months, so the surviving sample undercounts the relevant transactions. We test this directly by regressing community-quarter transaction counts and community-quarter south-facing shares

on the community-quarter average heat measure. Within community-quarter cells, the variation our design uses, both coefficients are positive and statistically significant (+0.370, $t = 9.29$; +0.041, $t = 8.27$), the opposite sign of the prediction under heat-driven dropout of south-facing units. (The raw cross-sectional correlation runs the other way, reflecting that hotter cells are systematically different communities and quarters, which the fixed effects absorb.) If anything, transaction volume rises with the local heat measure within community-quarter cells, which makes the result on the interaction even harder to explain as a selection artifact.

Alternative orientation definitions. The baseline south-facing dummy collapses "south" and "south-north" into a single category, following Lu (2018); the great majority (about 91 percent) of south-facing units are listed as pure south. Re-estimating with east-facing or west-facing units as the focal orientation gives interaction coefficients that are also negative (Table 12), consistent with the fact that east and west rooms also receive direct sunlight at parts of the day in summer; only due-north reverses sign, which is what the direct-solar-radiation channel requires. The relative magnitudes are coherent with a direct-solar-radiation story: south, east, and west each receive direct sunlight at some part of the day in summer and all three carry a measurable heat penalty, while north receives no direct sunlight at noon in Shanghai's latitude and reverses sign in the joint regression.

8 Welfare Analysis

To gauge the economic magnitude of the premium loss, take a representative apartment of 90 square meters at 50,000 yuan per square meter, with a total price $\bar{P} = 4.5$ million yuan. At zero heat exposure the south-facing premium is 1.61 percent, or 72,450 yuan. At the sample mean exposure of $\bar{H} = 0.4782$ (per-100-day units), the premium is 1.33 percent, or 59,706 yuan. The present-value heat penalty at mean exposure is therefore $|\beta_3| \cdot \bar{H} \cdot \bar{P} \approx 12,700$ yuan. To annualize this over a hold of T years at discount rate r , we divide by the standard annuity factor

$$A(r, T) = \frac{1 - (1 + r)^{-T}}{r}, \quad (11)$$

which yields an implied annual cost of $|\beta_3| \cdot \bar{H} \cdot \bar{P} / A(r, T)$. With $T = 30$ years and $r = 0.04$, $A(0.04, 30) = 17.292$, and the annual cost is roughly 736 yuan. Scaling instead from a single typical Shanghai summer of about sixty P90 days gives an annual penalty in the order of fifteen thousand yuan in present value, or near nine hundred yuan per year on the same annuity basis.

The 736-yuan-per-year estimate is in the same range as the energy-cost component reported in [Deschênes and Greenstone \(2011\)](#), who find that residential energy expenditure responds substantially to extreme-temperature days. [Zhang et al. \(2022\)](#) use the China Residential Energy Consumption Survey and report that each additional day with a mean temperature above 32°C raises household electricity consumption by 8.9 percent relative to a day in the 10 to 16°C range, with the response driven by more intensive air-conditioner use. [Auffhammer \(2022\)](#) obtains a comparable adaptive electricity response in U.S. data. Both responses point to a measurable monetary cost of additional cooling on hot days, which is the part of the heat penalty buyers can in principle compute.

The premium loss we recover should be read as a sum of two components. The first is the discounted stream of higher summer cooling expenditures that fall on a south-facing unit, which the energy literature can price. The second is the non-monetary value of summer comfort, which is harder to quantify but consistently large in field measurements of thermal sensation ([Chen et al., 2015](#); [Lin and Deng, 2008](#)). Plausible values for the second component, taken from a range of Shanghai-based comfort studies, sit in the low-thousand-yuan-per-year range and place the total in line with typical Shanghai household summer electricity spending. The figure is best read as a monetary equivalent that makes the magnitude tangible; the variation we document across windows reflects how much weight recent heat carries at the point of pricing, not a precise resolution of any household’s future cost. The calibration is not meant to deliver a precise welfare estimate. A fuller welfare exercise would need building-level energy data and a model of air-conditioner adoption, which is beyond the scope of this paper.

9 Conclusion

The south-facing premium in Shanghai’s resale market is not a fixed characteristic price. It declines with cumulative extreme-heat exposure during the listing window. The baseline drop is 0.59 percentage points per 100 P90 days, the dilution-corrected single-summer estimate is about ten times larger, and the penalty is present in the seller’s listing price as well as the transaction price, locating the adjustment on the seller side. Humidity amplifies the penalty; wind offsets it. Low-rise buildings show the strongest response and north-facing units reverse sign.

Three strands of the literature look different in light of these results. Hedonic studies of orientation have treated the premium as a fixed coefficient, but inside a single community and quarter it moves with the weather during the listing window. Climate-capitalization studies at the city-year level ([Kang et al., 2024](#)) or with a fifteen-day deal window ([Chen](#)

et al., 2023) average across periods of very different effective exposure; cutting our sample to the cleanest single-summer subset gives a coefficient an order of magnitude larger. The day-of-deal sentiment results in Hu and Lee (2020) and Busse et al. (2015) place the price effect at the moment of signing, but our listing-price evidence puts it weeks or months earlier, on the seller side.

The results carry two practical implications. For appraisers and assessors who rely on orientation as a fixed feature in valuation models, the premium should be treated as state-dependent rather than constant. Ignoring listing-period weather will overstate the orientation contribution in hot-summer windows and understate it in cool-summer windows, with errors that average out at the city level but can be material at the unit level. For buyers and sellers, the listing-price evidence suggests that the heat penalty is already priced in early in the window rather than being available to extract through end-of-window bargaining. Because the penalty is embedded in the listing price early in the window rather than emerging at the close, the orientation premium is best assessed against listing-window conditions rather than treated as a fixed feature.

The results also offer a way to reconcile two strands of the recent China climate-housing literature that have appeared to point in different directions. Kang et al. (2024) and Chen et al. (2025) find that extreme heat reduces city-level house prices, consistent with a demand-side capitalization story. Chen et al. (2023) finds that extreme heat in the fifteen days before a land auction raises bid prices, attributing the result to heat-induced cognitive impairment among bidders. Our setting matches Kang et al. (2024) in horizon and Chen et al. (2023) in granularity. The response we isolate operates over a months-long listing window, the Kang et al. (2024) horizon, but within a single physical attribute. The comparison is between south and non-south units in the same community and quarter, so a city-wide cognitive effect cannot be behind it: that kind of effect would move both orientations together. As climate-housing research moves to finer spatial and temporal resolution, a within-attribute comparison like ours can help tell cumulative cost capitalization apart from effects that act at the moment of transaction.

Three limitations. The platform’s identifier is at the listing level, not the property level, so a true repeat-sales design is not feasible; the quasi-repeat-sales fixed effect we use (community by bedrooms by floor band by age band by area bin) gives a smaller but same-signed coefficient. The sample is Shanghai. The orientation-by-heat mechanism is physical and should apply to other mid-latitude Chinese cities, but the magnitude depends on local climate and building stock. A placebo on newly constructed units, where buyers cannot visit in advance, would sharpen the interpretation and is left to future work. Comparable exercises in northern Chinese cities, where the winter comfort margin is similarly visible, would also

be informative, as would extensions that use household-level energy data to discipline the welfare magnitudes in Section 8.

Declaration on the Use of Generative AI

During the preparation of this work, the authors used Anthropic’s Claude and OpenAI’s ChatGPT as assistive tools in two areas. First, both tools were consulted during code development for the Stata regression scripts, the Python data-cleaning pipeline, and the R figure code; their role was limited to suggesting syntax, debugging error messages, and refactoring long loops. All estimation logic, variable construction, sample restrictions, and the choice of specifications were designed and verified by the authors against the underlying Stata logs and raw data. Second, both tools were used during manuscript writing for language editing, restructuring of paragraphs, and stylistic revision. No section of the paper was generated end-to-end by an AI tool. The research design, the empirical findings, the interpretation of results, and the final wording of every sentence are the authors’ own. The authors have reviewed and edited the output of these tools and take full responsibility for the content of this publication.

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Appendix for “Extreme Heat and the Reduction of the South-Facing Premium: Evidence from Shanghai’s Housing Market”

(For Online Publication)

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Appendix A Tables

The orientation mechanism makes a mirror prediction for winter. A south-facing unit takes in more passive solar warmth on a cold day, so if the premium responds to orientation-specific comfort, cumulative cold during the listing window should raise the south-facing premium rather than lower it. Let $\text{Cold}_{i,t}$ be the count of days in the listing window with daily mean temperature below the local tenth percentile, scaled per 100 days as for heat. The joint specification is

$$\ln(P_{i,t}) = \alpha + \beta_1 S_i + \beta_2^H \text{Heat}_{i,t} + \beta_3^H S_i \times \text{Heat}_{i,t} + \beta_2^C \text{Cold}_{i,t} + \beta_3^C S_i \times \text{Cold}_{i,t} + X_i' \gamma + \delta + \varepsilon_{i,t}. \quad (\text{A1})$$

Entered on its own, the south-by-cold interaction is negative (-0.0257), because a window long enough to contain both hot and cold days is a long window, and over such windows the cold term stands in for the heat term. With both terms in the regression, the heat interaction is $\beta_3^H = -0.0128$ ($p < 0.01$) and the cold interaction turns positive, $\beta_3^C = +0.0619$ ($p < 0.01$): a south-facing unit loses value when the window is hot and gains value when it is cold. This is the sign pattern that an orientation-comfort channel predicts and that a flat “bad weather lowers the premium” story does not, and it complements the orientation placebo of Section 7,

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where only due-north reverses sign. The two point in the same direction from opposite ends: the penalty tracks the orientations that take in direct summer sun, and the winter bonus tracks the orientation that benefits from it in the cold.

We read this winter result as suggestive rather than decisive, for a reason worth stating plainly. The positive cold interaction appears only in the joint specification, which is identified from windows long enough to contain both an extreme-heat tail and an extreme-cold tail. These are exactly the multi-season windows that Section 7 flags as diluted, and summer heat and the following winter’s cold may themselves be linked through regional climate, so the joint design cannot separate the two as cleanly as the single-summer evidence in the main text. We therefore do not lean on the magnitude of β_3^C , and treat it as a consistency check on the direction of the mechanism rather than as a second main estimate.

This result and the heating-degree-day result in Table 5 measure different things, which is why their signs differ. The HDD term sums over every day below the base and is dominated by the many mildly cool days in a window, which carry no orientation advantage; its interaction is small and weakly negative. The tenth-percentile count picks out only the genuinely cold tail of winter, when a south-facing room’s passive warmth is worth the most, and there the sign turns positive. Because hot and cold days occur at very different frequencies, the per-hundred-day coefficients are not comparable; the standardised ones are. In standardised units, one additional standard deviation of summer-heat exposure lowers the south-facing premium by 0.67 percentage points, while one standard deviation of winter-cold exposure raises it by 0.36 percentage points. Summer heat is the larger of the two, as expected at Shanghai’s latitude.

TABLE A1 Summary Statistics for Control Variables

Variable	Obs.	Mean	Std. Dev.	Min	Max
Building area (m^2)	226,188	79.00	34.09	29.50	216.58
Building area squared	226,188	7,402.63	6,758.68	870.25	46,906.90
Building age	224,550	18.72	9.82	0	98
Building age squared	224,550	446.82	482.06	0	9,604
Bedrooms	229,763	1.97	0.82	0	9
Living rooms	229,763	1.35	0.63	0	8
Bathrooms	229,763	1.18	0.47	0	7
High floor	230,783	0.37	0.48	0	1
Middle floor	230,783	0.36	0.48	0	1
Elevator	230,783	0.42	0.49	0	1
Fine decoration	230,783	0.26	0.44	0	1
Simple decoration	230,783	0.16	0.37	0	1
Rough decoration	230,783	0.03	0.18	0	1
Slab building	230,783	0.92	0.27	0	1
Tower building	230,783	0.08	0.27	0	1
Steel-concrete structure	230,783	0.56	0.50	0	1
Brick-concrete structure	230,783	0.43	0.50	0	1
Single-level layout	230,783	0.40	0.49	0	1
Duplex layout	230,783	0.00	0.05	0	1
70-year property right	230,783	0.25	0.44	0	1
50-year property right	230,783	0.00	0.06	0	1
Number of residential units	228,596	1,514.91	1,179.37	1	8,172
Number of buildings	228,643	31.40	33.36	1	549
Greening rate (%)	227,084	35.78	9.39	0.15	100.00
Property management fee	227,248	1.42	2.68	0.08	223.00
Floor area ratio	226,976	2.18	3.16	0.01	100.00
School district	230,763	0.49	0.50	0	1

Notes: This table reports summary statistics for the housing and neighborhood control variables used in the baseline regressions. Binary variables are coded as indicators taking values of 0 or 1.